HDM Pairwise Comparison Calculator

☐ Categorical information (check to enter the perspectives; uncheck to enter criteria for one perspective)

Mission

Cyber Readiness

Calculate | Respondent ID 65 | Add to cumulative table

Perspective for these criteria

E: Leader

Criteria for perspective E: Leader

Number of criteria

6

Descriptions (text), then pairwise comparisons: type the left score (1 to 99); the right score is 100 minus it.

A	Cybersecurity goals	**B**	Cybersecurity learning	**C**	Cybersecurity risk
D	Policies	**E**	Professionals	**F**	Supply Chain

A Cybersecurity goals	70	**B** Cybersecurity learning	30
B Cybersecurity learning	10	**C** Cybersecurity risk	90
C Cybersecurity risk	80	**D** Policies	20
D Policies	50	**E** Professionals	50
E Professionals	50	**F** Supply Chain	50

A Cybersecurity goals	20	**C** Cybersecurity risk	80
B Cybersecurity learning	40	**D** Policies	60
C Cybersecurity risk	70	**E** Professionals	30
D Policies	60	**F** Supply Chain	40

A Cybersecurity goals	50	**D** Policies	50
B Cybersecurity learning	50	**E** Professionals	50
C Cybersecurity risk	70	**F** Supply Chain	30

A Cybersecurity goals	60	**E** Professionals	40
B Cybersecurity learning	60	**F** Supply Chain	40

A Cybersecurity goals	60	**F** Supply Chain	40

Output

```
6 variables, 720 permutations
item    average score    std dev
  A        0.14774       0.03034
  B        0.08518       0.01905
  C        0.42270       0.05674
  D        0.12827       0.02658
  E        0.11527       0.02464
  F        0.10085       0.02231
HDM Inconsistency (mean std dev):            0.02994
Square root of summed variances (RSV lookup): 0.07946
RSV table (6 variables): value falls below the lowest tabulated percentile (0.01)
```

Cumulative tables

Clear this table

Perspectives

ID	**A** Organizational	**B** Info Mgmt	**C** Sys Mgmt	**D** Professional	**E** Leader	Inconsistency	Remove
2	0.14	0.16	0.13	0.17	0.40	0.01	Remove
1	0.25	0.19	0.21	0.08	0.27	0.01	Remove
Mean	**0.20**	**0.17**	**0.17**	**0.13**	**0.33**		
Minimum	0.14	0.16	0.13	0.08	0.27		
Maximum	0.25	0.19	0.21	0.17	0.40		
Std. Deviation	0.08	0.02	0.05	0.06	0.09		
Disagreement						0.009	

Criteria of A: Organizational

ID	**A** Computer users settings and	**B** Cybersecurity Readiness Assessments	**C** Documents are marked and protected	**D** Presence of Implementaon Oversight	**E** Presence of legislave understanding	**F** Social impact of breaches is talked about in	**G** There is an organizational common vocabulary	Inconsistency	Remove

	permissions are known					the company			
22	0.33	0.17	0.29	0.09	0.04	0.04	0.04	0.04	Remove
23	0.16	0.15	0.14	0.14	0.19	0.09	0.13	0.03	Remove
Mean	**0.24**	**0.16**	**0.21**	**0.11**	**0.11**	**0.07**	**0.09**		
Minimum	0.16	0.15	0.14	0.09	0.04	0.04	0.04		
Maximum	0.33	0.17	0.29	0.14	0.19	0.09	0.13		
Std. Deviation	0.12	0.01	0.11	0.03	0.11	0.03	0.06		
Disagreement								0.038	

Criteria of B: Info Mgmt

ID	**A** Info Officer is in contact with Internet Svc Provider	**B** Logging is sufficient for security and forensics	**C** Network and System admin procedures documented	**D** Retention periods are in place and used	**E** Standards are understood	Inconsistency	Remove
38	0.04	0.29	0.39	0.13	0.16	0.05	Remove
1	0.07	0.19	0.25	0.15	0.32	0.06	Remove
Mean	**0.06**	**0.24**	**0.32**	**0.14**	**0.24**		
Minimum	0.04	0.19	0.25	0.13	0.16		
Maximum	0.07	0.29	0.39	0.15	0.32		
Std. Deviation	0.03	0.07	0.10	0.02	0.12		
Disagreement						0.054	

Criteria of C: Sys Mgmt

ID	**A** Data loss	**B** Energy system outages	**C** Machine Limitations	**D** Network Modeling	**E** Outages not required	**F** Planning for forensic	Inconsistency	Remove
555	0.22	0.22	0.08	0.02	0.25	0.21	0.06	Remove
1000	0.11	0.18	0.11	0.17	0.24	0.17	0.01	Remove

Mean	**0.16**	**0.20**	**0.10**	**0.09**	**0.25**	**0.19**		
Minimum	0.11	0.18	0.08	0.02	0.24	0.17		
Maximum	0.22	0.22	0.11	0.17	0.25	0.21		
Std. Deviation	0.07	0.03	0.02	0.11	0.01	0.03		
Disagreement							0.036	

Criteria of D: Professional

ID	**A** Cyber awareness	**B** External Reporting	**C** External vendor	**D** Threats considered	**E** Change Mgmt	Inconsistency	Remove
65	0.24	0.14	0.15	0.31	0.16	0.02	Remove
Mean	**0.24**	**0.14**	**0.15**	**0.31**	**0.16**		
Minimum	0.24	0.14	0.15	0.31	0.16		
Maximum	0.24	0.14	0.15	0.31	0.16		
Std. Deviation							
Disagreement						0.016	

Criteria of E: Leader

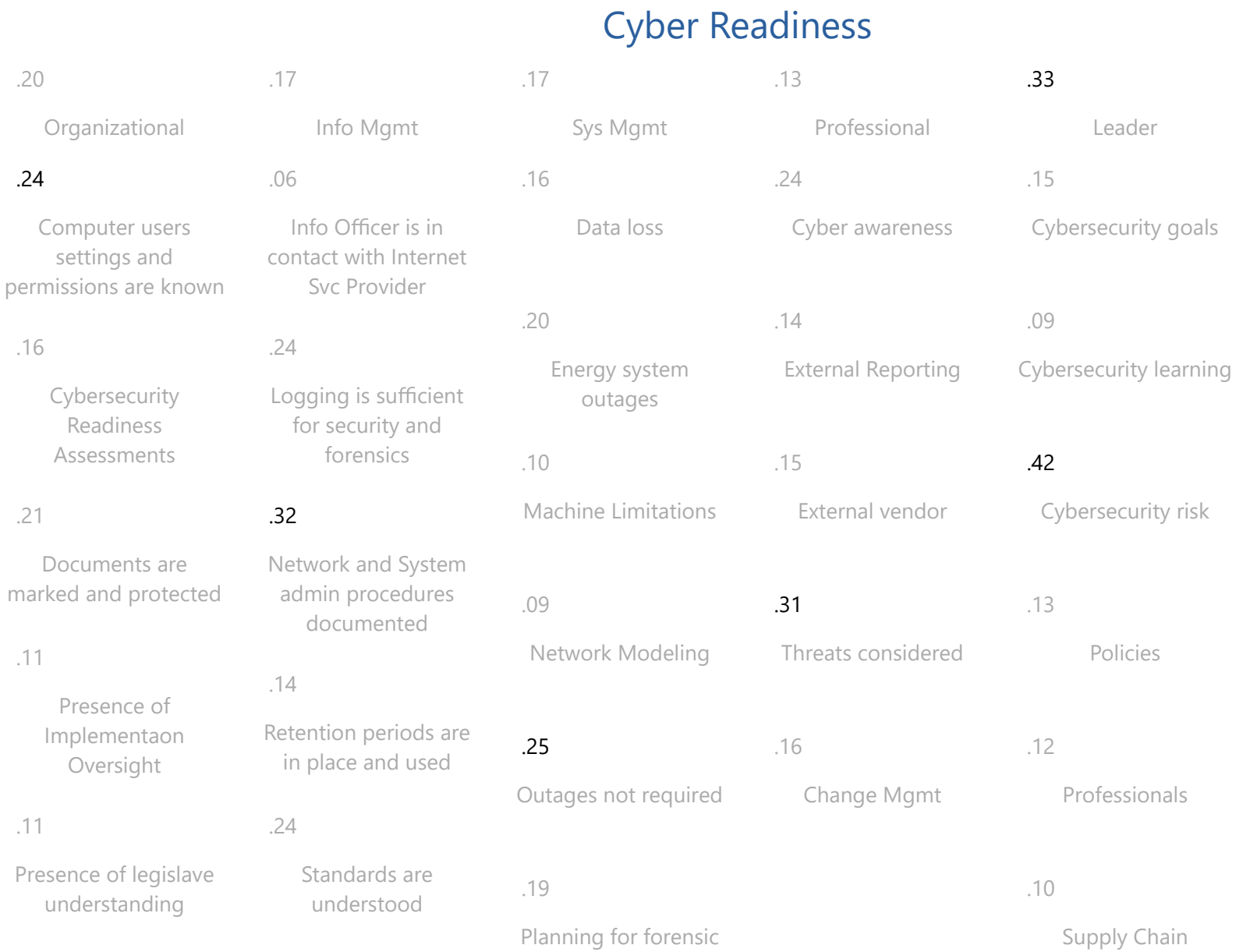
ID	**A** Cybersecurity goals	**B** Cybersecurity learning	**C** Cybersecurity risk	**D** Policies	**E** Professionals	**F** Supply Chain	Inconsistency	Remove
65	0.15	0.09	0.42	0.13	0.12	0.10	0.03	Remove
Mean	**0.15**	**0.09**	**0.42**	**0.13**	**0.12**	**0.10**		
Minimum	0.15	0.09	0.42	0.13	0.12	0.10		
Maximum	0.15	0.09	0.42	0.13	0.12	0.10		
Std. Deviation								
Disagreement							0.030	

Global weights

Perspectives	Value	Criterion	Local Weights	Global Weights
Organizational	0.2	Computer users settings and permissions are known	0.24	0.05
		Cybersecurity Readiness Assessments	0.16	0.03
		Documents are marked and protected	0.21	0.04
		Presence of Implementaon Oversight	0.11	0.02
		Presence of legislave understanding	0.11	0.02
		Social impact of breaches is talked about in the company	0.07	0.01
		There is an organizational common vocabulary	0.09	0.02
Info Mgmt	0.17	Info Officer is in contact with Internet Svc Provider	0.06	0.01
		Logging is sufficient for security and forensics	0.24	0.04
		Network and System admin procedures documented	0.32	0.06
		Retention periods are in place and used	0.14	0.02
		Standards are understood	0.24	0.04
Sys Mgmt	0.17	Data loss	0.16	0.03
		Energy system outages	0.2	0.04
		Machine Limitations	0.1	0.02
		Network Modeling	0.09	0.02
		Outages not required	0.25	0.04
		Planning for forensic	0.19	0.03
Professional	0.13	Cyber awareness	0.24	0.03
		External Reporting	0.14	0.02
		External vendor	0.15	0.02
		Threats considered	0.31	0.04
		Change Mgmt	0.16	0.02
Leader	0.33	Cybersecurity goals	0.15	0.05
		Cybersecurity learning	0.09	0.03
		Cybersecurity risk	0.42	0.14
		Policies	0.13	0.04

		Professionals	0.12	0.04
		Supply Chain	0.1	0.03
Total	**1**		**5**	**1**

Hierarchy chart



.07

Social impact of breaches is talked about in the company

.09

There is an organizational common vocabulary

Disagreement = average of the respondents' Inconsistency values. Global weight = perspective value x local weight, using the Mean rows of the cumulative tables.